

YISHUN JUNIOR COLLEGE PRELIMINARY EXAMINATION

CHEMISTRY
Higher 2

9746/01
25 AuG 2008
1 hr

Paper 1 Multiple Choice

Additional materials: ORS
Data Booklet



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name and CTG on the ORS form and shade your register number.

There are **40** questions on this paper. Answer **all** questions. For each question there are 4 possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the ORS.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this question paper.

You may use a calculator.

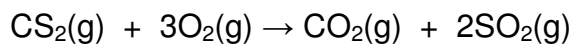
This question paper consists of **20** printed pages

Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1 Naturally occurring silicon is a mixture of three isotopes, ^{28}Si , ^{29}Si and ^{30}Si . The relative abundance of ^{29}Si and ^{30}Si is 7.8%.
The relative abundance of ^{28}Si is
- A** 91.1%
B 92.2%
C 95.0%
D 96.3%

- 2 Carbon disulphide, CS_2 , is a volatile flammable liquid used in the manufacture of cellophane. On combustion, CS_2 is oxidized as follows:



A 20 cm^3 sample of CS_2 vapour is ignited with 150 cm^3 of oxygen. The final volume of gases after burning is treated with an excess of aqueous alkali at room temperature. What is the number of molecules absorbed by aqueous alkali?

- A** 5.01×10^{20}
B 7.53×10^{20}
C 1.50×10^{21}
D 3.77×10^{21}

- 3 Ions of the two common isotopes of the transition metal nickel are shown below:



Which statement is **correct** ?

- A** ${}_{28}^{58}\text{Ni}^{2+}$ ion has greater number of nucleons in its nucleus than the ${}_{28}^{60}\text{Ni}^{2+}$ ion.
- B** ${}_{28}^{60}\text{Ni}^{2+}$ ion has more protons in its nucleus than the ${}_{28}^{58}\text{Ni}^{2+}$ ion.
- C** Both ions have an outer electronic configuration $3d^8 4s^0$.
- D** Both ions have more electrons than neutrons.
- 4 When 20g of solid Group II nitrate, $\text{X}(\text{NO}_3)_2$ was decomposed by heat, 0.135mol of its oxide, XO was formed. Which nitrate was heated ?
- A** Magnesium
- B** Calcium
- C** Strontium
- D** Barium

- 5 The table gives the successive ionisation energies for an element **X**.

	1 st	2 nd	3 rd	4 th	5 th
Ionisation energy kJmol^{-1}	580	1800	2700	11600	14800

What could be the formula of the compound formed when X react with chlorine?

- A** XCl
- B** XCl_2
- C** XCl_3
- D** XCl_4

6 Which molecule has the smallest dipole?

- A $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$
- B $\text{CH}_3\text{CHCNCH}_2\text{CH}_3$
- C $\text{CH}_2\text{NH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
- D $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$

7 When 25 cm^3 of 1 mol dm^{-3} potassium hydroxide is neutralized with an equal volume of 1.0 mol dm^{-3} hydrochloric acid, the temperature of the mixture rose by 6.8°C .

What will be the temperature change if 50 cm^3 of 0.5 mol dm^{-3} potassium hydroxide is neutralized with an equal volume of 0.5 mol dm^{-3} hydrochloric acid? (assume heat losses to be negligible in each case)

- A 1.7°C
- B 3.4°C
- C 6.8°C
- D 13.6°C

8 A current is passed through two voltmeters connected in series. The first voltmeter contains $\text{XSO}_4(\text{aq})$ while the second voltmeter contains $\text{Y}_2\text{SO}_4(\text{aq})$.

Which statement about this experiment is **correct**?

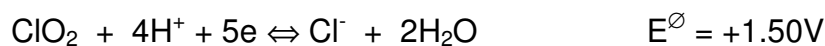
- A Equal masses of X and Y are deposited.
- B An equal number of moles of X and Y is deposited.
- C The number of moles of Y deposited is half the number of moles of X deposited.
- D The volume of oxygen gas collected in first voltmeter is the same volume collected in the second voltmeter, measured under the same conditions of temperature and pressure.

- 9 An ion of metal X can be oxidised by potassium manganate(VII) in acid solution to form XO_3^- . In an experiment, 1.25×10^{-3} mol of the ion of X required 37.5 cm^3 of $0.0200 \text{ mol dm}^{-3}$ potassium manganate(VII) for complete reaction.

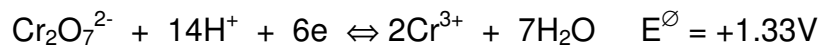
What is the initial oxidation state of the ion of X?

- A +1
- B +2
- C +3
- D +4

- 10 The following shows the standard redox potential for ClO_2 in acidic solution:



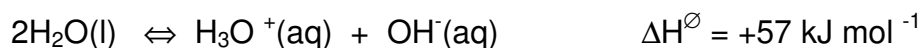
Some other standard redox potentials are given below:



Which statement is **correct**?

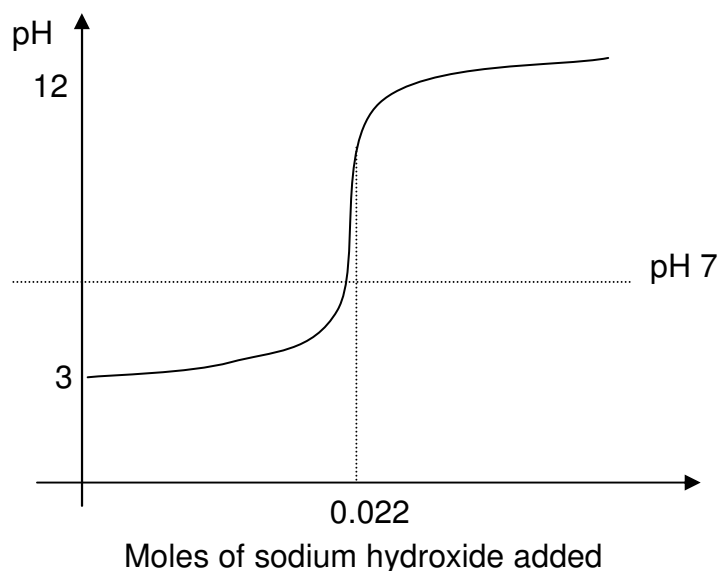
- A ClO_2 and chlorine gas react together to form chloride ions.
- B ClO_2 cannot oxidize bromide ions to bromine.
- C When ClO_2 is added to an aqueous solution of Cr^{3+} , Cr^{3+} becomes oxidized to $\text{Cr}_2\text{O}_7^{2-}$.
- D $\text{Cr}_2\text{O}_7^{2-}$ can oxidize chloride ions to chlorine gas but not ClO_2 .

- 11 Water dissociates into ions according to the following equilibrium:



Which statement is **incorrect** as temperature rises?

- A pH < 7.
 - B The dissociation constant increases.
 - C Water dissociates to a larger extent.
 - D The concentration of $\text{H}^+(\text{aq})$ is higher than that of $\text{OH}^-(\text{aq})$.
- 12 The pH change when aqueous sodium hydroxide is added dropwise to a solution of a weak acid is shown below.



When 0.022 moles of sodium hydroxide is used, pH of the resulting solution is greater than 7.

Which statement explains this observation?

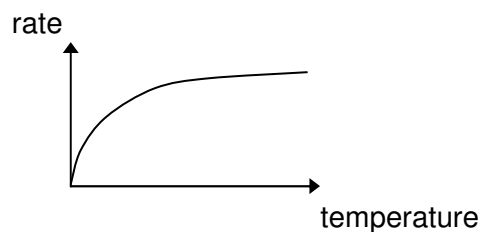
- A Conjugate base of the weak acid underwent hydrolysis.
- B Too much of sodium hydroxide is added.
- C A buffer solution is formed.
- D Neutralisation is incomplete.

- 13** One important buffer that exists in blood composed of $\text{H}_2\text{PO}_4^- (\text{aq})$ and HPO_4^{2-} . The net ionic equation that represents the reaction of hydrochloric acid with this buffer is

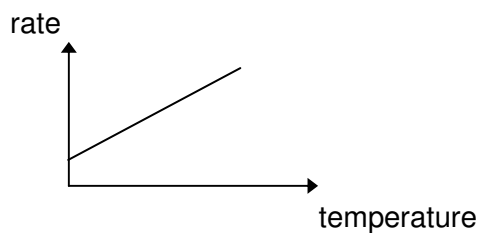
- A** $\text{HCl}(\text{aq}) + \text{H}_2\text{PO}_4^- (\text{aq}) \rightleftharpoons \text{Cl}^- (\text{aq}) + \text{H}_3\text{PO}_4(\text{aq})$
B $\text{H}_3\text{O}^+(\text{aq}) + \text{HPO}_4^{2-}(\text{aq}) \rightleftharpoons \text{H}_2\text{O}(\text{l}) + \text{H}_2\text{PO}_4^-(\text{aq})$
C $\text{H}_3\text{O}^+(\text{aq}) + \text{H}_2\text{PO}_4^-(\text{aq}) \rightleftharpoons \text{H}_2\text{O}(\text{l}) + \text{H}_3\text{PO}_4(\text{aq})$
D $\text{HCl}(\text{aq}) + \text{HPO}_4^{2-}(\text{aq}) \rightleftharpoons \text{Cl}^- (\text{aq}) + \text{H}_2\text{PO}_4^-(\text{aq})$

- 14** Which of the following graphs describes the kinetics of an enzyme-catalysed reaction against temperature?

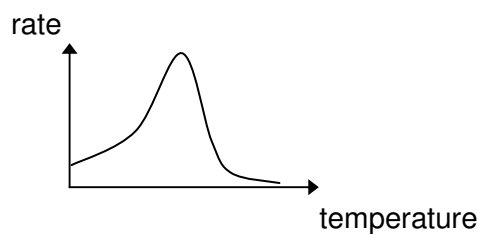
A



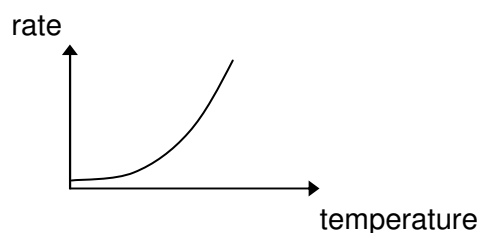
B



C

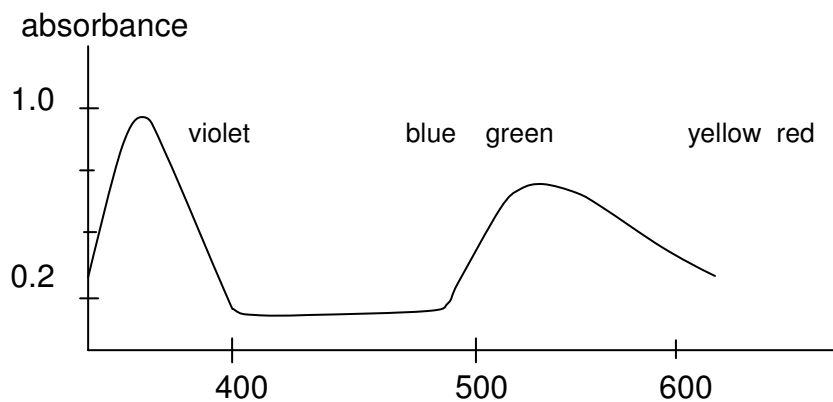


D



- 15 Which statement shows a decreasing trend from left to right across the elements of the third period of the Periodic Table?
- A The radii of the atoms.
 - B Melting points of the elements.
 - C First ionization of the elements.
 - D Electrical conductivity of the elements.
- 16 Which statement is **not correct** for the hydrogen halides of Group VII, HCl, HBr and HI?
- A Aqueous HCl is the weakest acid.
 - B Thermal stability decreases from HCl to HI.
 - C HI is the strongest reducing agent.
 - D Bromine can displace chlorine from a solution of HCl.
- 17 Which statement best explains why Mg reacts slowly with cold water and yet barium reacts readily with cold water?
- A Mg has higher first and second ionization energies than Ba.
 - B The standard reduction potential of Mg is less negative than that of Ba.
 - C The polarizing power of Mg^{2+} is greater than that of Ba^{2+} .
 - D Mg^{2+} has a more exothermic hydration energy than that of Ba^{2+} .

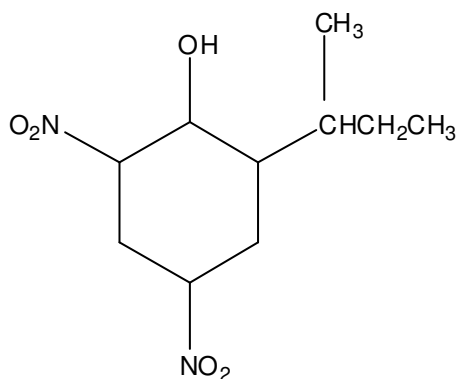
- 18 The diagram shows the uv/visible spectrum of aqueous transition metal complex ion, $[\text{M}(\text{H}_2\text{O})_6]^{3+}(\text{aq})$



On oxidation of $[\text{M}(\text{H}_2\text{O})_6]^{3+}$ to $[\text{M}(\text{H}_2\text{O})_6]^{4+}$, the solution turned colourless. Deduce the electronic configuration of the transition metal **M**.

- A $[\text{Ar}] 3\text{d}^1 4\text{s}^2$
- B $[\text{Ar}] 3\text{d}^2 4\text{s}^2$
- C $[\text{Ar}] 3\text{d}^3 4\text{s}^2$
- D $[\text{Ar}] 3\text{d}^{10} 4\text{s}^2$

- 19 Binapacryl is used as a fungicide.



What deduction about Binapacryl can be made from this structure?

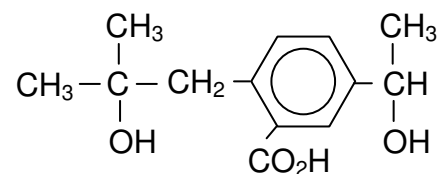
- A It reacts readily with iodine in aqueous sodium hydroxide.
 - B It's solution is acidic.
 - C It contains a tertiary alcohol.
 - D It gives a ketone when oxidized.
- 20 A compound **X** displays optical activity. One mole of **X** evolves two moles of hydrogen when it reacts with excess sodium.
- What could be the formula of **X** ?

- A $\text{CH}_2(\text{OH})\text{CH}(\text{OH})\text{CH}(\text{OH})_2$
- B $\text{HOOCCH}_2\text{CH}_2\text{OH}$
- C $\text{CH}(\text{OH})_2\text{CH}_2\text{CH}(\text{OH})\text{CH}(\text{OH})_2$
- D $\text{C}_6\text{H}_5\text{COOCH}_2\text{OH}$

21 Which pairs of compounds is the boiling point of the first compound **higher** than the second compound?

- A $\text{CH}_3\text{CH}(\text{CO}_2\text{H})_2$ and $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$
- B $\text{HOCH}(\text{CH}_2\text{CH}_2\text{CO}_2\text{H})\text{CO}_2\text{H}$ and $\text{HOOCCH}(\text{CH}_2\text{CH}_2\text{CO}_2\text{H})\text{CO}_2\text{H}$
- C $(\text{CH}_3)_2\text{CHCH}(\text{NH}_2)\text{CONH}_2$ and $(\text{CH}_3)_2\text{CHCH}(\text{NH}_2)\text{COOH}$
- D $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$ and $\text{H}_2\text{NCH}_2\text{CH}_2\text{NHCH}_2\text{CH}_2\text{NH}_2$

22 The compound **ibuprofen** is a powerful painkiller.



Which statement about the compound is **not correct** ?

- A Hydrogen bonds formed by ibuprofen are stronger than O-H covalent bonds.
- B All the carbon atoms in benzene in ibuprofen are sp^2 hybridised.
- C All the bonds surrounding each carbon in CH_3 are 109.5° .
- D There are 2 lone pairs of electrons in each oxygen atom of the hydroxyl groups.

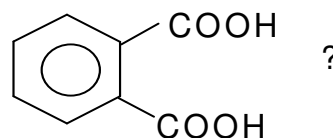
23 Which compound is stable when concentrated aqueous sodium hydroxide is added?

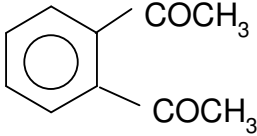
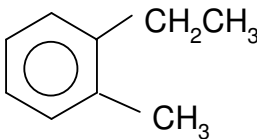
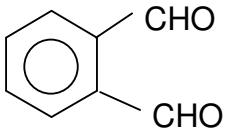
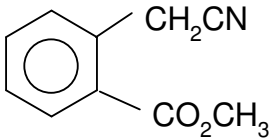
- A $\text{CH}_3\text{CH}_2\text{CONHCH}_3$
- B $\text{CH}_3\text{CH}_2\text{COOCH}_3$
- C $\text{CH}_3\text{CH}_2\text{CN}$
- D $\text{CH}_3\text{CH}_2\text{C}_6\text{H}_5$

24 Ethane reacts with chlorine gas in the presence of ultraviolet light to form a mixture of products via free radical substitution. Which statement is **correct** about this reaction?

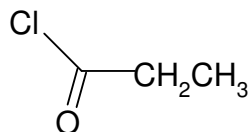
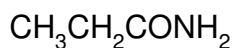
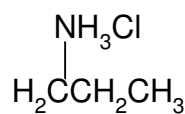
- A Homolytic fission occurs only at initiation step.
- B Butane is present in small quantities in the product mixture.
- C Bond formation occurs only in the termination step.
- D Chloroethane is formed only in the propagation step.

25 Which of the following reactions would **not** produce



- A  + alkaline aqueous iodine followed by acidification
- B  + acidified KMnO_4 , reflux
- C  + aqueous Tollen's reagent, warm, followed by acidification
- D  + aqueous H_2SO_4 , reflux

- 26 Equimolar aqueous solutions of each of the following compounds are made.

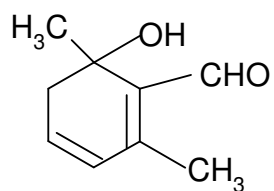
**X****Y****Z**

What would be the correct order of increasing pH?

- A** Z, Y, X
B Y, Z, X
C X, Y, Z
D X, Z, Y
- 27 The planet Jupiter is known to have a reducing atmosphere consisting mainly of methane and ammonia, with some ethane and water in the form of ice crystals. Astronomers used the impact of fragments of Comet Shoemaker-Levy with Jupiter to identify more complex molecules present in the planet's atmosphere. Which molecule is present?

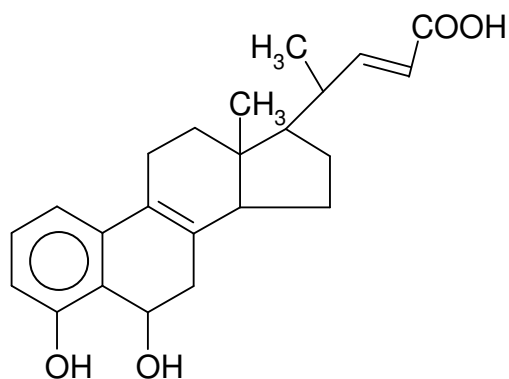
- A** $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
B $\text{C}_6\text{H}_5\text{OH}$
C $\text{CH}_3\text{CH}_2\text{NH}_2$
D $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$

28 What are the products of warming **J** with Tollen's reagent?



- A** Silver mirror and a carboxylic acid.
- B** Silver mirror and a carboxylate salt.
- C** Silver mirror, a ketone and a carboxylic acid.
- D** Silver mirror, a ketone and a carboxylate salt.

29 How many chiral carbons are present in the molecule shown below?



- A** 3
- B** 4
- C** 5
- D** 6

30 Which salt will be the most basic in aqueous solution?

- A** $\text{ClCH}_2\text{CH}_2\text{COO}^-\text{K}^+$
- B** $\text{FCH}_2\text{CH}_2\text{COO}^-\text{K}^+$
- C** $\text{C}_6\text{H}_5\text{NH}_3^+\text{Cl}^-$
- D** KNO_3

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

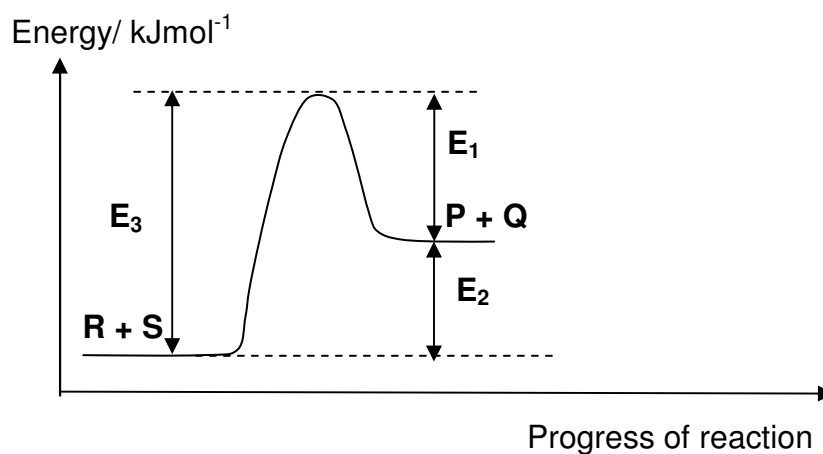
- 31** The ion T^{5+} contains 43 protons and 46 neutrons.
What deduction can be made from this information about T^{5+} ?
- 1** The third ionization energy is significantly higher than the second ionization energy.
 - 2** The electronic configuration of T^{5+} is $[\text{Kr}] 4d^2$.
 - 3** The aqueous solution of T^{5+} is coloured.
- 32** A gas is believed to behave ideally and is described exactly by the equation $pV=nRT$. It is true that
- 1** such a gas cannot be liquefied.
 - 2** the pressure of a fixed mass of such a gas is inversely proportional to the volume at constant temperature.
 - 3** there is intermolecular force between particles of such a gas.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 33** The diagram below shows the energy profile for the following reversible reaction:



It can be inferred from the diagram that

- 1** the addition of a catalyst would alter the value of E_2 .
- 2** the backward reaction is exothermic.
- 3** increasing temperature does not affect the value of E_3 and favours the forward reaction.

The response **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 34** Barium reacts with cold water to form barium hydroxide and hydrogen.



The standard enthalpy change of this reaction can be measured in the laboratory. This reaction has been found to occur spontaneously at room temperature.

What information is needed to conclude the spontaneity of this reaction?

- 1** There is an increase in entropy.
- 2** The reaction is exothermic.
- 3** $\Delta G^\ominus$ of the reaction is negative.

- 35** The table shows data about two elements, sodium and copper.

Element	Atomic Number	Melting point/$^\circ\text{C}$
Sodium	11	97.8
Copper	29	1083

Which of the following can be deduced from the given data ?

- 1** Magnesium has a melting point between 97.8°C and 1083°C .
- 2** Copper has greater number of delocalized electrons.
- 3** Melting point decrease with increasing atomic number.

The response **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

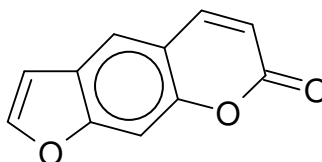
No other combination of statements is used as a correct response.

36 Use of the Data Booklet is relevant to this equation.

Which of the following is/are chemically unstable when left to stand in air?

- 1** A solution of potassium hexacyanoferrate(III).
- 2** A solution of iron(II) chloride.
- 3** A mixture of aqueous sodium hydroxide and iron(II) sulphate.

37 Bergaptol belongs to the family of furanocoumarins which are used medically to promote skin pigmentation. It has the structure shown below:



Which of the following is/are **not correct** about Bergaptol ?

- 1** It can react with a maximum of 3 moles of bromine in tetrachloromethane in the dark.
- 2** It is unstable in both acid and basic medium.
- 3** It decolourises an aqueous solution of cold potassium manganate(VII).

The response **A** to **D** should be selected on the basis of

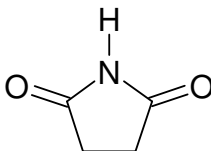
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 38** Compounds which may be synthesized directly from ethanoyl chloride by a nucleophilic substitution reaction include

- 1** CH_3COCH_3
- 2** CH_3COOH
- 3** CH_3CONH_2

- 39** Compound **X** has the structural formula as shown below:



Which of the following regarding **X** is/are **correct** ?

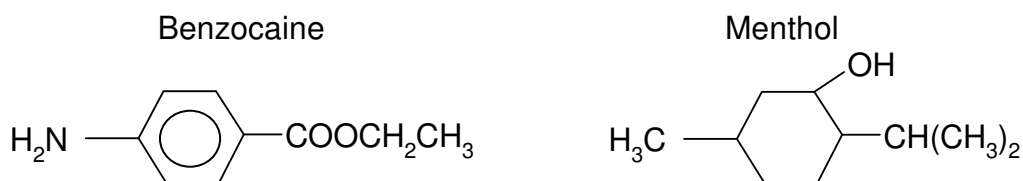
- 1** **X** reacts with chlorine gas in the presence of ultra-violet light.
- 2** **X** reacts with boiling aqueous sodium hydroxide with the evolution of ammonia.
- 3** **X** reacts with boiling hydrochloric acid to yield an amino acid.

The response **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 40** Benzocaine and menthol are local anesthetics used in sunburn ointments and skin lotions. Their structures are shown below:



Which of the following reagents and conditions can be used to distinguish between the two compounds?

- 1** Aqueous bromine, dark.
- 2** Acidified aqueous potassium manganate(VII), heat.
- 3** Iodine in aqueous sodium hydroxide, heat.

End of paper